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Gustatory Responsiveness to Six Bitter Tastants in Three Species of Nonhuman Primates

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Abstract Gustatory responsiveness of six adult squirrel monkeys, four spider monkeys, and five pigtail macaques to six bitter tastants was assessed in two-bottle preference tests of brief duration (2 min). Animals were given the choice between a 30-mM sucrose solution and defined concentrations of a bitter tastant dissolved in a 30-mM sucrose solution. With this procedure, *Saimiri sciureus*, *Ateles geoffroyi*, and *Macaca nemestrina* were found to significantly discriminate concentrations as low as 0.2, 0.05, and 0.1 mM quinine hydrochloride; 1, 1, and 0.05 mM caffeine; 20, 5, and 1 mM naringin; 5, 2, and 1 mM salicin; 0.01, 0.001, and 0.02 mM sucrose octaacetate; and 0.05, 0.01, and 0.5 mM denatonium benzoate, from the alternative stimulus. With the exception of naringin in the pigtail macaques, all three species rejected all suprathreshold concentrations of all bitter tastants tested. The spider monkeys and the pigtail macaques displayed the lowest taste avoidance thresholds with three of the six tastants each; in contrast, the squirrel monkeys displayed the highest taste avoidance thresholds with four of the six tastants. The across-tastant patterns of taste avoidance thresholds were identical in spider monkeys and squirrel monkeys; both species displayed the following order of sensitivity: sucrose octaacetate > denatonium benzoate > quinine hydrochloride > caffeine > salicin > naringin. All three primate species were more sensitive to the two artificial tastants (sucrose octaacetate and denatonium benzoate) compared to the four naturally occurring tastants. However, the concentrations detected by all three primate species with the four naturally occurring tastants are well below those reported in plants or arthropods consumed by these species suggesting that they may use bitterness as a criterion for food selection.

Keywords Taste preference thresholds · Bitter taste · *Saimiri* · *Ateles* · *Macaca*

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Introduction

The sense of taste has been investigated behaviorally (e.g., Glaser 1986; Pritchard et al. 1994; Hladik et al. 2002) and electrophysiologically (e.g., Hellekant and Ninomiya 1994; Scott et al. 1998, 1999) in a number of primate species. However, most studies have concentrated on detectability of the five basic taste qualities, mainly by using the alkaloid quinine hydrochloride as the only prototypic bitter stimulus. Only sparse information is available on the behavioral responsiveness of nonhuman primates to naturally occurring bitter tastants other than this alkaloid. This is all the more surprising given the wide variety of chemical classes that comprise naturally occurring bitter-tasting substances (DuBois 2007) and given that the vast majority of food-related toxins taste bitter to humans (Rouseff 1990). Further, both plants and arthropods frequently use bitter-tasting and often highly toxic compounds as deterrents against potential predators (Belitz and Wiester 1985; Eisner et al. 2005). Therefore, it should be adaptive for primates to detect bitter tastants at concentrations low enough to prevent ingestion of too high amounts of such potentially toxic compounds.

Given the presumed importance of bitter tastants for food selection in primates, it was the aim of the present

study to assess the gustatory responsiveness of three species of nonhuman primates to six bitter tastants belonging to different chemical classes. Four of these tastants, quinine hydrochloride, caffeine, naringin, and salicin, occur naturally in both plants and arthropods, whereas the other two, sucrose octaacetate and denatonium benzoate, have not been found so far in nature (Belitz and Wiester 1985).

As all primate species tested so far reject substances that taste bitter to humans, a two-bottle preference test of brief duration (2 min) was employed. This method makes it possible to measure directly preferences for or aversions to tastants and largely rules out the possibility of postingestive factors on the animals' ingestive behavior. Further, as all three primates used in the present study cooperated in tests of liquid consumption only as long as at least one of the alternatives was a sapid solution, we decided to use a 30-mM sucrose solution rather than water, both as the solvent for the tastant and as the alternative stimulus. This concentration of sucrose is only a factor of 3 above the taste preference threshold of the squirrel monkeys and the pigtail macaques, respectively, and a factor of 10 above that of the spider monkeys and thus represents a weak sweet stimulus that is unlikely to mask the bitter tastant.

The three primate species employed here, squirrel monkeys (*Saimiri sciureus*), spider monkeys (*Ateles geoffroyi*), and pigtail macaques (*Macaca nemestrina*), are known to differ—at least to some degree—in their dietary habits (Caldecott 1986; Lima and Ferrari 2003; Gonzalez-Zamora et al. 2009), thus allowing us to address the question whether possible differences in bitter taste responsiveness may reflect an evolutionary adaptation to dietary specialization. The ability to include two New World and one Old World primate species (the macaques) allowed us to address additionally the question whether the degree of phylogenetic relatedness rather than dietary specialization may affect the bitter taste responsiveness of nonhuman primates.

Methods and Materials

Animals Testing was carried out with six male adult squirrel monkeys (*S. sciureus*), two male and three female adult pigtail macaques (*M. nemestrina*), and four female adult spider monkeys (*A. geoffroyi*). Animals were housed as social groups in enclosures with adjacent single cages that could be closed by sliding doors to allow temporary separation of animals for individual testing (for details of maintenance, see Laska 1996, 2000; Laska et al. 1996). All animals were born in captivity and had no experience with natural vegetation. They were fed commercial monkey chow, fresh fruit, and vegetables ad libitum but were deprived of water overnight before testing on the following morning. The amount of food offered daily to the animals was such that leftovers still were present on the floor the next morning. Thus, it was unlikely that ravenous appetite affected the animals' ingestive behavior during the tests.

Taste Stimuli A set of six substances tasting bitter to humans was used: quinine hydrochloride (quinine is a naturally occurring alkaloid in the bark of cinchona trees; as it is a basic amine, it is presented as a salt, in this case as a hydrochloride, Chemical Abstracts Service # 130-89-2), caffeine (a naturally occurring xanthine alkaloid, e.g., in coffee plants, CAS# 58-08-2), naringin (a naturally occurring flavonoid glycoside, e.g., in grapefruit, CAS# 10236-47-2), salicin (a naturally occurring phenol glycoside in many Salicaceae, CAS# 138-52-3), sucrose octaacetate (an artificial acetylated sucrose derivative, CAS# 126-14-7), and denatonium benzoate (an artificial compound with the International Union of Pure and Applied Chemistry name phenylmethyl-[2-[(2,6-dimethylphenyl)amino]-2-oxoethyl]-diethylammonium benzoate, CAS# 3734-33-6). The rationale for using these substances was that they represent different chemical classes and are naturally occurring as well as artificial bitter tastants. All substances were obtained from Sigma-Aldrich (St. Louis, MO, USA) and were of the highest available purity.

Procedure Taste responsiveness to the six bitter tastants mentioned above was assessed by using a two-bottle preference test of short duration (Richter and Campbell 1940). Twice each day, approximately 2 and 1 h before feeding, the animals were separated and allowed 2 min to drink from a pair of simultaneously presented graduated cylinders with metal drinking spouts.

Animals were given a choice between a 30-mM sucrose solution and defined concentrations of the bitter tastants dissolved in a 30-mM sucrose solution. With quinine hydrochloride, caffeine, naringin, and salicin, testing started at a concentration of 10 mM and proceeded in the following steps (10, 5, 2, 1, 0.5, 0.2, 0.1 mM, etc.) until the animals failed to show a significant preference or aversion. With sucrose octaacetate and denatonium benzoate, testing started at a concentration of 1 mM and proceeded in the following steps (1, 0.5, 0.2, 0.1, 0.05 mM, etc.).

To maintain the animals' motivation and willingness to cooperate, testing did not follow a strict order of increasing or decreasing bitter tastant concentrations but followed a pseudorandomized scheme in which trials with taste mixtures containing high and thus presumably aversive concentrations of a bitter tastant were alternated with mixtures containing low and thus presumably less aversive concentrations of the same bitter tastant.

Each pair of stimuli was presented a total of ten times, and the position of the stimuli was pseudorandomized in

order to counterbalance possible position preferences. All animals had served in previous studies that use the same method (Laska and Hernandez Salazar 2004, Laska et al. 2008). They were trained to enter the single cages voluntarily and were completely accustomed to the procedure.

The experiments reported here comply with the *Guide for the Care and Use of Laboratory Animals* (National Institutes of Health Publication no. 86-23, revised 1985) and also with current German and Mexican laws.

Data Analysis For each animal, the amount of liquid consumed from each bottle was recorded, summed for the ten test trials with a given stimulus combination, converted to percentages (relative to the total amount of liquid consumed from both bottles), and 66.7% (that is, two thirds of the total amount of liquid consumed) was taken as criterion of preference. This rather conservative criterion was chosen for reasons of comparability of data, as the same criterion had been used in previous studies that used the same method with the same primate species (Laska 1994, 1996, 1997, 1999, 2000; Laska et al. 1996, 1998, 1999, 2000a, b, 2001, 2008; Laska and Hernandez Salazar 2004), and in order to avoid misinterpretation of data due to a too liberal criterion.

Additionally, two-tailed binomial tests (Siegel and Castellan 1988) were performed, and an animal was only regarded as significantly preferring one of the two alternative stimuli if it reached the criterion of 66.7% and consumed more from the bottle containing the preferred stimulus in at least eight out of ten trials (binomial test, $P< 0.05$). Accordingly, at the group level, a given species was regarded only as significantly preferring one of the two alternative stimuli if all animals tested reached the criterion of 66.7%, and all animals consumed more from the bottle containing the preferred stimulus in at least eight out of ten trials.

Preliminary tests showed that the animals generally rejected solutions containing relatively high concentrations of the bitter tastants. Following convention, the results nevertheless are expressed as percentage preference for the tastant and not for the solvent. Accordingly, 33.3% (that is, one third of the total amount of liquid consumed) was taken as criterion of aversion.

Preliminary analysis of the data indicated that there were no reliable differences in choice behavior and liquid consumption between males and females of a species and between the first and the second presentation of the day. Intraindividual variability of the amount of liquid consumed across the ten test trials with a given stimulus combination was low and averaged less than 20%. Thus, a theoretically possible bias in the overall preference score due to excessive drinking in aberrant trials did not occur. Therefore, data obtained in ten test trials from an individual animal of a given species with a given stimulus combination were combined and are reported as group means and standard deviations.

Results

Figures 1, 2, and 3 show the mean taste responses of squirrel monkeys, spider monkeys, and pigtail macaques, respectively, in the two-bottle preference tests with a 30-mM sucrose solution used both as solvent for the bitter tastants and as the alternative stimulus.

Squirrel monkeys significantly discriminated concentrations as low as 0.2 mM quinine hydrochloride, 1.0 mM caffeine, 20 mM naringin, 5 mM salicin, 0.01 mM sucrose octaacetate, and 0.05 mM denatonium benzoate from the alternative stimulus (group level). In most cases, interindividual variability of scores was low, as can be inferred from the SDs in Fig. 1. Thus, at the individual level, the avoidance threshold values for a given tastant maximally differed by a factor of 5 (quinine hydrochloride, naringin, and denatonium benzoate), and usually less, between subjects.

At the group level, the spider monkeys significantly discriminated concentrations as low as 0.05 mM quinine hydrochloride, 1.0 mM caffeine, 5 mM naringin, 2 mM salicin, 0.001 mM sucrose octaacetate, and 0.01 mM denatonium benzoate from the alternative stimulus. As for the squirrel monkeys, interindividual variability of scores was usually low (compare SDs in Fig. 2) and avoidance threshold values for a given tastant maximally differed by a factor of 5 (quinine hydrochloride), and in one case (caffeine) not at all, between subjects.

Pigtail macaques showed significant discrimination of concentrations as low as 0.1 mM quinine hydrochloride, 0.05 mM caffeine, 1 mM naringin, 1 mM salicin, 0.02 mM sucrose octaacetate, and 0.5 mM denatonium benzoate from the alternative stimulus (group level). Again, interindividual variability of scores was low in most cases (compare SDs in Fig. 3). Thus, at the individual level, the avoidance threshold values for a given tastant maximally differed by a factor of 5 (caffeine), and in two cases (quinine hydrochloride and denatonium benzoate) not at all, between subjects. Interestingly, the pigtail macaques preferred rather than rejected naringin at concentrations between 10 and 1 mM.

With all six bitter tastants and all three species, we observed that, in each trial, each animal sampled both alternatives at least once and, with the exception of the highest bitter tastant concentrations, even tended to switch between bottles more than once.

Comparison Among Species Among the three primates tested, the spider monkeys (with quinine hydrochloride,

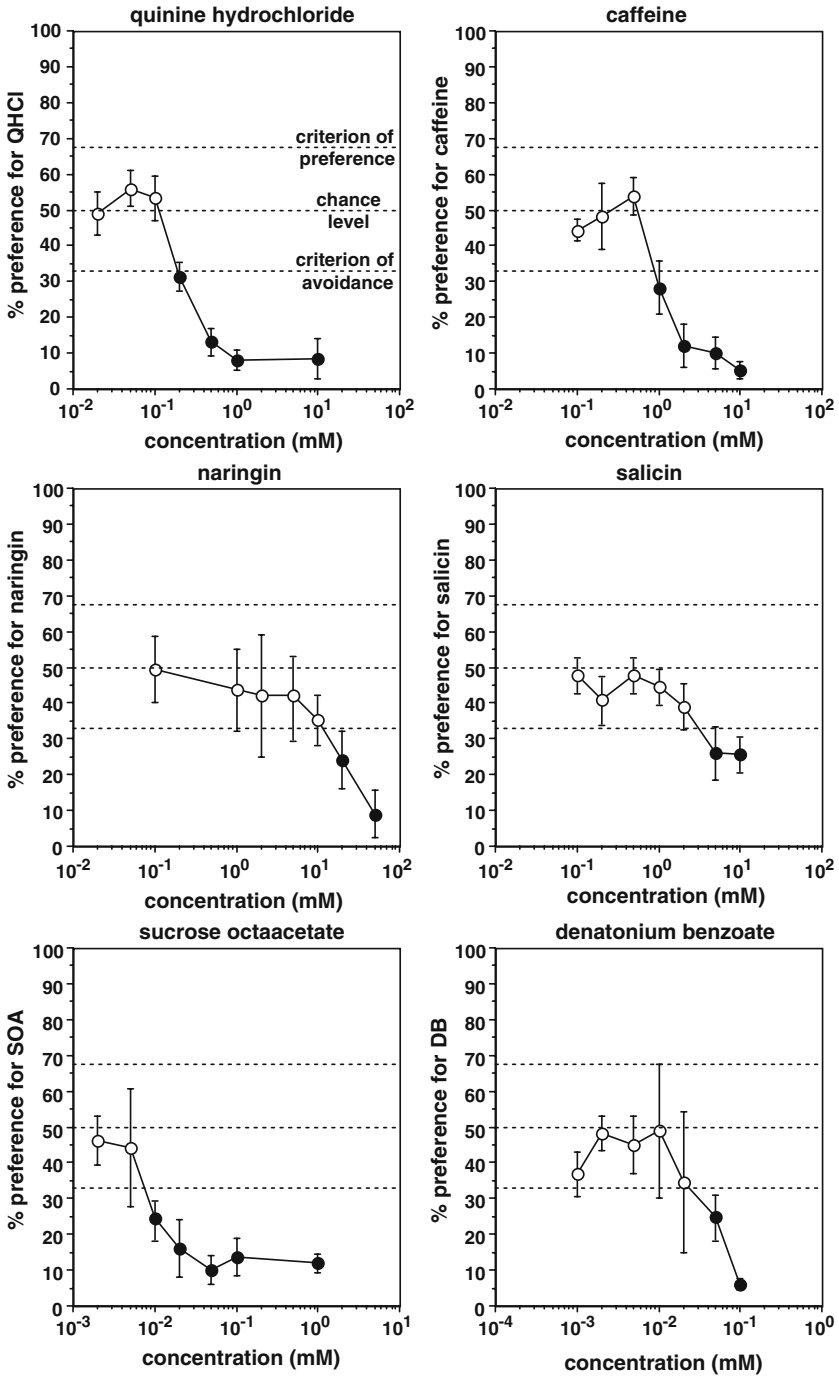


Fig. 1 Gustatory responsiveness of six squirrel monkeys to various concentrations of quinine hydrochloride, caffeine, naringin, salicin, sucrose octaacetate, and denatonium benzoate dissolved in a 30-mM sucrose solution and tested against a 30-mM sucrose solution as an alternative stimulus. Each data point represents the mean value (±SD) of ten test sessions of 2 min per animal. *Black* data points indicate concentrations at which all animals significantly rejected the bitter tastant (binomial test, $P<0.05$)

sucrose octaacetate, and denatonium benzoate) and the pigtail macaques (with caffeine, naringin, and salicin) showed the lowest taste avoidance thresholds with three of the six tastants each. The squirrel monkeys, in contrast, showed the highest taste avoidance thresholds with four of the six tastants (quinine hydrochloride, caffeine, naringin, and salicin). However, the across-tastant patterns of taste avoidance thresholds were identical between spider monkeys and squirrel monkeys, with both species displaying the following order of sensitivity: sucrose octaacetate > denatonium benzoate > quinine hydrochloride > caffeine > salicin > naringin. Accordingly, a significant positive correlation between the across-tastant patterns of sensitivity for the tastants tested with these two species was found (Spearman, $r_s=0.76$, $P<0.01$). The pigtail macaques displayed an across-tastant pattern of taste avoidance thresholds (sucrose octaacetate > caffeine > quinine hydrochloride > denatonium benzoate > salicin = naringin), which differed from that of the two other species. Accordingly, no significant correlation between the across-tastant pattern of *M.*

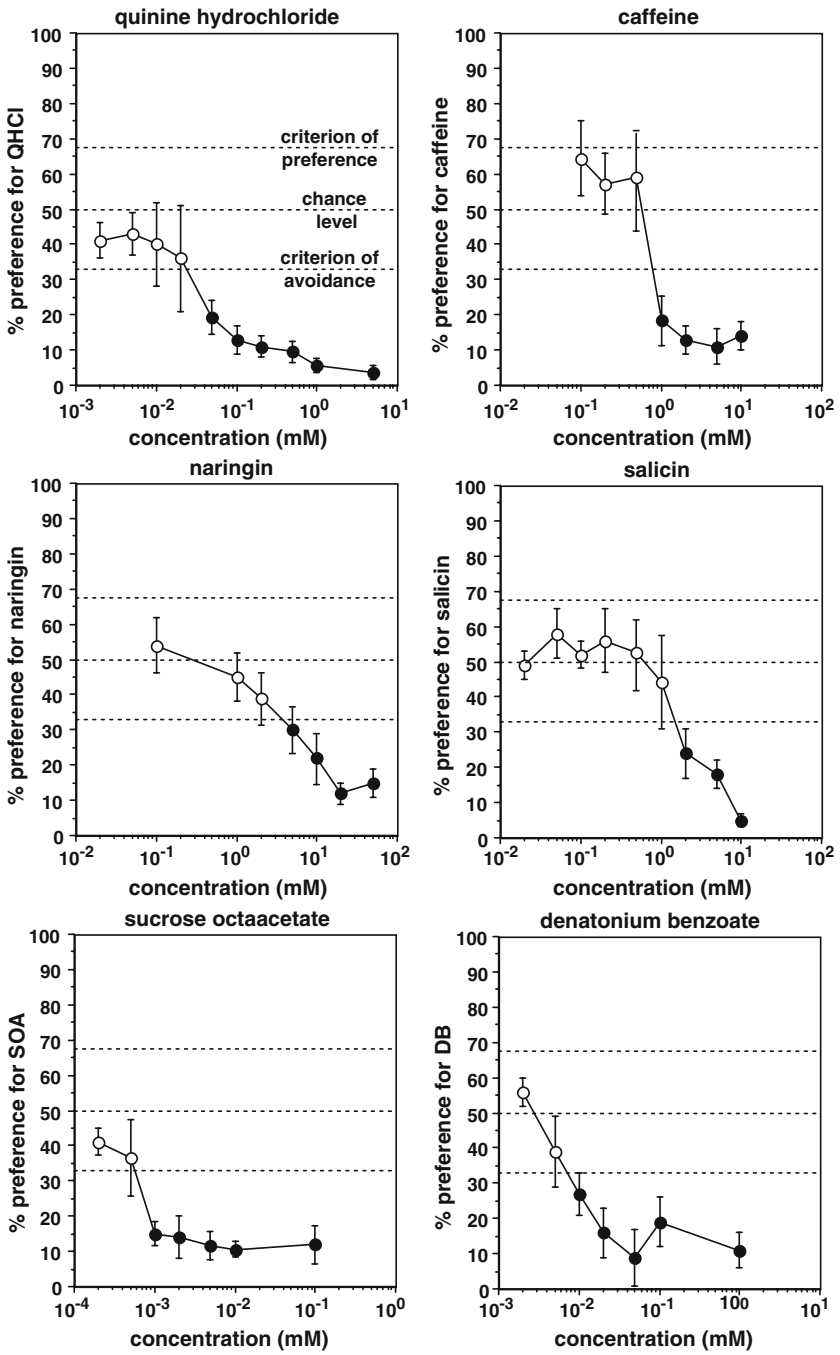


Fig. 2 Gustatory responsiveness of four spider monkeys to various concentrations of quinine hydrochloride, caffeine, naringin, salicin, sucrose octaacetate, and denatonium benzoate dissolved in a 30-mM sucrose solution and tested against a 30-mM sucrose solution as an alternative stimulus. Each data point represents the mean value (±SD) of ten test sessions of 2 min per animal. *Black* data points indicate concentrations at which all animals significantly rejected the bitter tastant (binomial test, $P<0.05$)

nemestrina and of *A. geoffroyi* (Spearman, $r_s=0.24$, $P>0.05$) and *S. sciureus* (Spearman, $r_s=0.26$, $P>0.05$) was found. Nevertheless, all primate species were most sensitive to sucrose octaacetate and least sensitive to naringin.

Discussion

The results of this study are a first and conservative approximation of the gustatory sensitivity of three primate species for the six bitter tastants tested. Although only between four and six animals per species were tested, the results appear robust as interindividual variability generally was low (see SDs in Figs. 1, 2, and 3). Further, with all tastants and all species, the lowest concentrations presented were consumed at equal amounts compared to the alternative stimulus, suggesting that the avoidance of (or, in the case of naringin with the pigtail macaques, preference for) higher concentrations of a stimulus was based on taste perception and not on other cues. However, the possibility of the tastants acting on olfactory and/or trigeminal receptors in the nose and/or the oral cavity cannot be

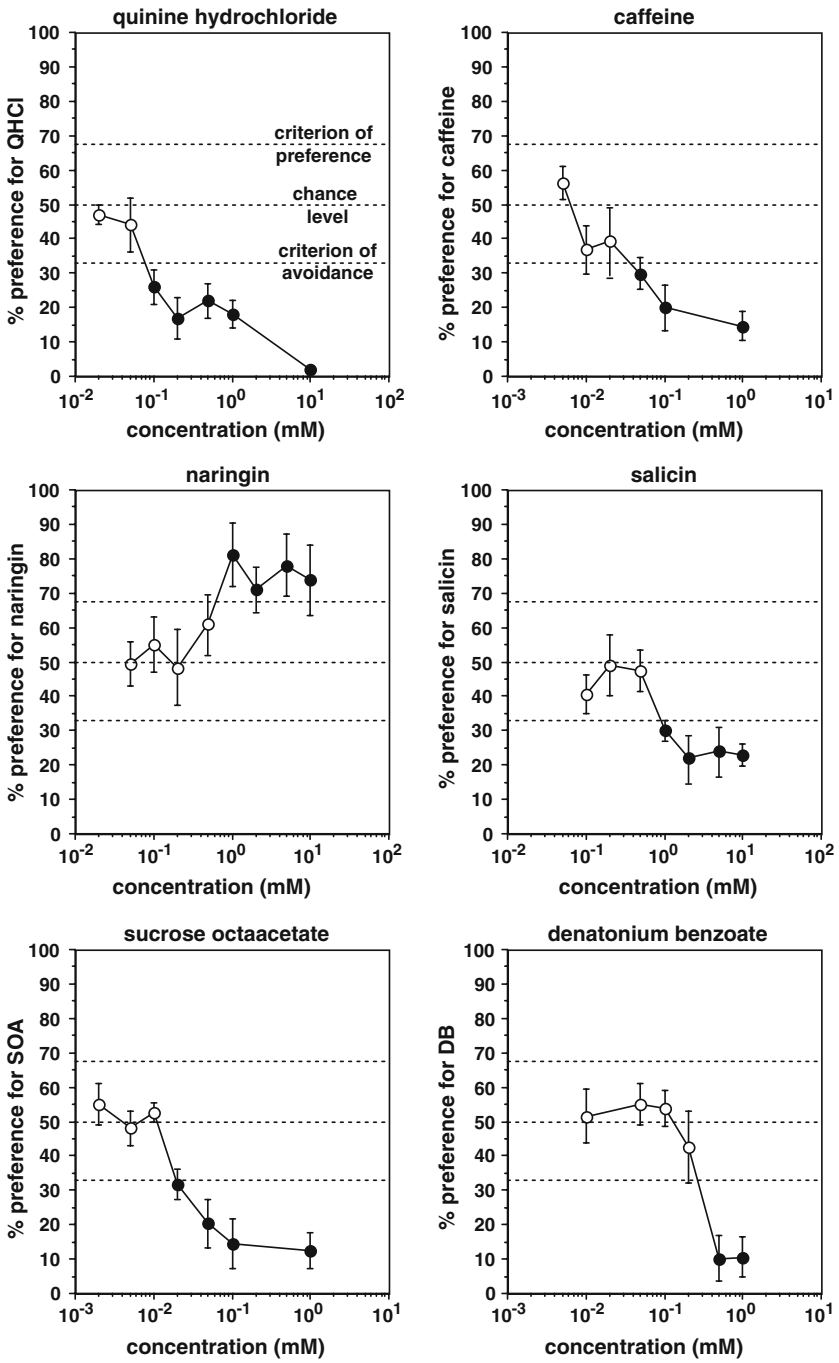


Fig. 3 Gustatory responsiveness of five pigtail macaques to various concentrations of quinine hydrochloride, caffeine, naringin, salicin, sucrose octaacetate, and denatonium benzoate dissolved in a 30-mM sucrose solution and tested against a 30-mM sucrose solution as an alternative stimulus. Each data point represents the mean value (±SD) of ten test sessions of 2 min per animal. *Black* data points indicate concentrations at which all animals significantly rejected (or, in the case of naringin, preferred) the bitter tastant (binomial test, $P<0.05$)

excluded. Similarly, the possibility that postingestive factors might have led to a conditioned aversion across trials that, in turn, might have affected the results cannot be excluded. There was no indication that the animals' avoidance of the bitter stimuli changed across trials. Taste qualities such as "bitter" are a human construct; it is not known whether other species experience the same percept as we do. Nevertheless, there is evidence of nonhuman primate taste receptors specialized to respond to and taste fibers, as well as cortical areas specialized to convey and process information about chemicals described as "bitter" by humans (Hellekant and Ninomiya 1994; Scott et al. 1999; Behrens and Meyerhof 2006).

Table 1 compares the taste avoidance thresholds of *S. sciureus*, *A. geoffroyi*, and *M. nemestrina* obtained here with data available from other species. Such across-species comparisons should take into consideration that different methods may lead to differing results. The present study employed a weak sucrose solution as solvent for the tastants, whereas other studies employed water. Similarly, in our study, the stimuli were presented for 2 min per trial, whereas some others have used different presentation times.

Additionally, the criterion used to define an avoidance threshold may differ among studies. All studies included in Table 1, however, employed two-bottle preference tests and were not conditioned to taste aversion or operant conditioning procedures and thus are basically the same method.

Whereas taste responsiveness to quinine hydrochloride has been tested with more than a dozen species of nonhuman primates and an even higher number of nonprimate mammals, data for the other tastants used in this study are sparse. The taste avoidance threshold of spider monkeys for quinine hydrochloride (0.05 mM) is one of the lowest values among the nonhuman primate species tested. It is interesting that the taste avoidance threshold of pigtail macaques for quinine hydrochloride (0.1 mM) is a factor 6–8 lower than that of the rhesus macaque and a factor of 10 lower than that of the long-tailed macaque, all species belonging to the same genus *Macaca*. All primate species tested here compare favorably in their responsiveness to quinine hydrochloride with the mouse. Other nonprimate mammals such as the cat (0.005 mM), the pig (0.002 mM), and the European mole (0.0005 mM) appear to be more sensitive for quinine hydrochloride than the primates tested here, whereas the horse (5.0 mM) and the rabbit (2.0 mM) show markedly higher avoidance threshold values and appear to be less sensitive than the primates tested here.

The only other data for any of the five other tastants used in the present study with other species of nonhuman primates were obtained with the rhesus macaque (Aspen et al. 1999). This Old World primate seems to be markedly less sensitive for caffeine (1.5 mM) than the pigtail macaque (0.05 mM) but similar to spider monkeys and squirrel monkeys (1.0 mM, respectively) and slightly more sensitive to denatonium benzoate (0.22 mM) than the pigtail macaque (0.5 mM), but less sensitive than the spider monkey (0.01 mM) and the squirrel monkey (0.05 mM).

The only species of nonprimate mammals for which data from all six tastants are at hand is the mouse. This rodent species appears to be less sensitive for caffeine and salicin and more sensitive for naringin than the three primates, but with avoidance threshold values for quinine hydrochloride, sucrose octaacetate, and denatonium benzoate falling into the range obtained with the three primate species. Although the across-tastant pattern of avoidance thresholds in the mouse (sucrose octaacetate > denatonium benzoate = quinine hydrochloride > naringin > caffeine = salicin) differs from those of the nonhuman primates tested, this species, too, appears to be more sensitive to the two artificial tastants compared to the four naturally occurring ones.

Our finding that the taste avoidance thresholds found here were about one (caffeine, naringin, and salicin) to four (denatonium benzoate) log units higher than the taste detection thresholds established in humans suggests that *Homo sapiens* is markedly more sensitive to bitter tastants than the primates tested here and, in fact, any nonhuman primate species tested so far (see Table 1). However, the sophisticated psychophysical signal detection methods employed with human subjects are likely to be more sensitive than the simple two-bottle preference test used with both nonhuman primates and nonprimate mammals. This assumption is supported by findings from Pritchard et al. (1995) who reported that taste thresholds obtained with a conditioning paradigm in *Macaca mulatta* are 1.6–2.2 log units lower than those obtained using a preference paradigm with the same species and are approximately 1.5 log units lower than human detection thresholds. This suggests that the sensitivity of the primates employed in our study might—at least for some of the tastants tested—be similar to or even match that of *H. sapiens*. The across-tastant pattern of taste detection thresholds obtained in humans (denatonium benzoate > sucrose octaacetate > quinine hydrochloride > caffeine > naringin > salicin) differs from those of the primates tested in our study, but humans, too, appear to be more sensitive to the two artificial bitter tastants compared to the naturally occurring ones.

Which factors might cause the differences in responsiveness within and between the species tested? It is well established that postingestive factors may influence an animal's ingestive behavior in subsequent encounters with the same type of food or liquid (Garcia and Hankins 1975). The method employed here, however, makes this possibility unlikely as the short duration of the tests (2 min per presentation) prevented animals from drinking amounts of liquid that are large enough to induce physiological effects that are perceptible.

Differences in dietary habits repeatedly have been shown to explain differences in taste performance among species (Kare 1971; Pfaffmann 1977; Spector 2000). Among New World primates, for example, the degree of frugivory correlates positively with sensitivity for food-associated monosaccharides and disaccharides (Laska 1996; Laska et al. 1996). Similarly, the proportion of animal matter in the diet of nonhuman primates correlates negatively with sensitivity for monosodium glutamate (Laska and Hernandez Salazar 2004). A comparison of 30 mammal species found a relationship between sensitivity for quinine hydrochloride and dietary specialization, with carnivores > omnivores > grazers > browsers (Glendinning 1994). This finding is consistent with the idea that animals with a relatively high occurrence of bitter and potentially toxic compounds in their diet, such as browsing herbivores, should have evolved a high bitter taste threshold (i.e., a low sensitivity) and, thus, a high tolerance to dietary toxins. In contrast, animals that rarely encounter bitter and potentially toxic compounds, such as carnivores, should have evolved a low bitter taste threshold (i.e., a high sensitivity) and, thus, a low tolerance to dietary toxins (Iason and Villalba 2006).

Table 1 Taste avoidance thresholds (in millimolar) for the six bitter tastants in the three primate species tested here (bold) and in other primates and nonprimate mammal species

species	QHCl	Caf	Nar	Sal	SOA	DB
Ateles geoffroyi	0.05	1.0	5.0	2.0	0.001	0.01
Saimiri sciureus	0.2	1.0	20	5.0	0.01	0.05
Macaca nemestrina	0.1	0.05	1.0	1.0	0.02	0.5
Homo sapiens	0.00097 (Glaser 1986)	0.002 (Schiffman et al. 1994)	0.22 (Schiffman et al. 1994)	0.2 (Bufe et al. 2002)	0.00036 (Schiffman et al. 1994)	0.0000042 (Schiffman et al. 1994)
	0.00048 (Schiffman et al. 1994)	3.2 (Smagghe and Louis-Sylvestre 1998)	0.14 (Smagghe and Louis-Sylvestre 1998)	0.2 (Soldo and Hofmann 2005)	0.004 (Boughter and Whitney 1993a, b)	
		1.0 (Soldo and Hofmann 2005)	0.14 (Drewnowksi et al. 1997)			
Macaca mulatta	0.63 (Glaser 1986)	1.5 (Aspen et al. 1999)				0.22 (Aspen et al. 1999)
	0.77 (Aspen et al. 1999)					
Macaca fascicularis	1.0 (Pritchard et al. 1994)					
Pan troglodytes	0.16 (Glaser 1986)					
Cheirogaleus medius	0.8 (Glaser 1986)					
Microcebus murinus	0.8 (Glaser 1986)					
	0.8 (Iaconelli and Simmen 2002)					
Loris tardigradus	0.8 (Glaser 1986)					
Nycticebus coucang	0.39 (Glaser 1986)					
Galago senegalensis	0.39 (Glaser 1986)					
Aotus trivirgatus	1.56 (Glaser 1986)					
Cebuella pygmaea	0.8 (Glaser 1986)					
	0.73 (Simmen and Hladik 1998)					
Saguinus midas niger	0.05 (Glaser 1986)					
Saguinus oedipus	0.065 (Simmen and Hladik 1998)					
Callithrix jacchus	0.53 (Simmen and Hladik 1998)					
Callithrix geoffroyi	0.37 (Simmen and Hladik 1998)					
Leontopithecus rosalia	0.21 (Simmen and Hladik 1998)					
Leontopithecus chrysomelas	0.055 (Simmen and Hladik 1998)					
Callimico goeldii	0.08 (Simmen and Hladik 1998)					

Nonprimate mammals						
Mus musculus	0.1 (Glendinning et al. 2002)	10 (Bachmanov, personal communication)	0.5 (Bachmanov, personal communication)	10 (Bachmanov, personal communication)	0.003 (Boughter et al. 2002)	0.1 (Whitney and Harder 1994)
					0.001 (Gannon and Whitney 1989)	0.2 (Boughter et al. 2005)
	0.2 (Boughter et al. 2005)				0.0001 (Harder et al. 1984)	0.1 (Bachmanov, personal communication)
					0.002 (Whitney and Harder 1994)	
Rattus norvegicus	0.27 (Brasser et al. 2005)	1.0 (Tordoff et al. 2008)			0.19 (Stewart et al. 1994)	0.55 (Brasser et al. 2005)
	0.02 (Cicala and McMichael 1964)					
	0.012 (Koh and Teitelbaum 1961)					0.032 (Tordoff et al. 2008)
	0.032 (Tordoff et al. 2008)					
	0.025 (Young et al. 1963)					
Clethrionomys glareolus	0.0093 (Chrzanowski 1965)					
Peromyscus melanotis	0.1 (Glendinning 1993)				1.0 (Glendinning 1993)	
Octodon degus	0.36 (Chrzanowski 1965)					
Mesocricetus auratus	2.0 (Carpenter 1956)	3.0 (Frank et al. 2004)			1.0 (Frank et al. 2004)	1.0 (Frank et al. 2004)
	0.3 (Frank et al. 2004)					
Cavia porcellus					1.0 (Jacobs 1978)	
Erethizon dorsatum	0.31 (Bloom et al. 1973)					
Felis silvestris catus	0.005 (Carpenter 1956)					
Oryctolagus cuniculus	2.0 (Carpenter 1956)					
Talpa europaea	0.0005 (Chrzanowski 1965)					
Crocidura russula	0.005 (Chrzanowski 1965)					
Erinaceus europaeus	0.1 (Ganchrow 1977)					
Odocoileus hemionus	0.13 (Crawford and Church 1971)					
Ovis aries	0.008 (Crawford and Church 1971)					
	0.48 (Goatcher and Church 1970a)					
	0.12 (Goatcher and Church 1970b)					
Sus scrofa	0.002 (Glaser et al. 2001)					0.1 (Nelson and Sanregret 1997)
Capra hircus	0.016 (Goatcher and Church 1970b)					
Bos taurus	0.06 (Goatcher and Church 1970b)					
Equus caballus	5.0 (Randall et al. 1978)					

QHCl quinine hydrochloride, *Caf* caffeine, *Nar* naringin, *Sal* salicin, *SOA* sucrose octaacetate, *DB* denatonium benzoate

This idea is supported further by the fact that browsing herbivores that cannot afford to reject all foods that taste bitter usually have evolved physiological mechanisms that aid in detoxifying potential toxins upon ingestion, while carnivores are usually lacking such evolutionary adaptations in their gastrointestinal tracts (Belovsky and Schmitz 1991).

The three species tested in this study differ markedly in the proportion of animal matter in their diet. However, squirrel monkeys, the species showing the highest taste avoidance thresholds with the naturally occurring tastants, also have been reported to show the highest proportion of animal matter in their diet (72% of total intake) among the three species tested. Pigtail macaques, the species with the lowest avoidance threshold with three of the four naturally occurring tastants, on the other hand, include a considerably lower proportion of animal matter in their diet (13%). Spider monkeys, with an intermediate position in their bitter taste responsiveness, hardly consume any animal matter (1% of total intake; Caldecott 1986; Lima and Ferrari 2003; Gonzalez-Zamora et al. 2009). Thus, our findings do not seem to support the presumed positive correlation between bitter taste sensitivity and carnivory. However, animal matter consumed by squirrel monkeys consists mainly of arthropods rather than vertebrates. Arthropods frequently employ bitter-tasting secretions or hemolymph as deterrents against predators (Eisner et al. 2005). The relatively low bitter taste sensitivity observed in squirrel monkeys might be explained as an evolutionary adaptation to bitter-tasting arthropod deterrents. This idea is supported by the fact that the pigtail macaques, which displayed lower avoidance thresholds for quinine hydrochloride than two other members of the genus *Macaca*, include a lower proportion of arthropods in their diet than rhesus and long-tailed macaques (Caldecott 1986).

Old World and New World primates differ in their ability to perceive substances such as aspartame and thaumatin, which both taste sweet to humans, suggesting that phylogenetic relatedness also might account for differences or correspondences in taste perception among species (Nofre et al. 1996). However, in our study, spider monkeys and pigtail macaques, i.e., one New World and one Old World primate species, displayed the lowest taste avoidance thresholds with three of the six tastants. The possibility of phylogenetic relatedness as an explanation for the observed differences in responsiveness to the bitter tastants tested is unlikely.

Pigtail macaques did not reject detectable concentrations of naringin but rather preferred them over the alternative stimulus. Although our use of a 30-mM sucrose solution, both as the solvent for the tastants and as the alternative stimulus, might have contributed to this unusual pattern of performance, this seems unlikely, as this concentration of sucrose is only a factor of 3 above this species' preference threshold (Laska 2000) and thought to be weak enough to prevent massive masking effects or other types of qualitative interactions that may occur when applied in heterogenous taste mixtures (Stevens 1996). A more likely explanation for the observed preference of the pigtail macaques for naringin is that this substance may not have a purely bitter taste for this species but may evoke a mixed taste sensation involving a quality that is not bitter. Human psychophysical studies (Schiffman 2000) as well as electrophysiological recordings from single taste fibers in nonhuman primates (Hellekant and Ninomiya 1994) have demonstrated that a variety of substances evoke mixed taste sensations and elicit activity in more than one type of taste fibers, i.e., not only in bitter-best fibers.

The concentrations detected by the three primate species with the four naturally occurring tastants are well below those reported in fruits or other parts of plants that are consumed by these species (Nagy and Shaw 1980; Souci et al. 1989; Rouseff 1990). However, bitter tastants may not be distributed uniformly in plants but may be concentrated in specific regions of fruits, seeds, or leaves. Nevertheless, it seems reasonable to assume that the bitterness elicited by these compounds in the natural diet of *S. sciureus*, *A. geoffroyi*, and *M. nemestrina* plays a part in the food selection behavior of these species.

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